Prabhavit Gupta

Fullstack Engineer | Generative AI, Agent Frameworks, Advanced RAG & Resilient Backend Systems

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Executive Summary

- **Generative AI Agent Architectures & Frameworks:** Applied AI software engineer with 3 years commercial experience building, evaluating, and operating autonomous AI agent frameworks and tool loops. Hands-on experience developing modular agent frameworks with LangGraph, LangChain, and Python state machines for complex multi-step reasoning and autonomous execution.
- **Advanced RAG & Robust Retrieval Engineering:** Deep practical expertise architecting high-throughput Retrieval-Augmented Generation (RAG) pipelines, semantic chunking, dense vector embeddings, and hybrid search on PostgreSQL pgvector. Experienced building security-first API integrations and custom Model Context Protocol (MCP) tool connectors.
- **Agent Evaluation, Tracing & Defensive Engineering:** Proven track record implementing systematic evaluation harnesses (measuring agent decision faithfulness, tool selection accuracy, and hallucination prevention using RAGAS), distributed tracing (Langfuse / OpenTelemetry), and containerized microservices in fast-paced startup environments.

Technical Skills

- **Agent Frameworks & Generative AI:** Autonomous AI Agents, Tool Loops, LangGraph, LangChain, ReAct Reasoning, Multi-Agent Coordination, Model Context Protocol (MCP Server & Client), Prompt Engineering, Structured Outputs (Pydantic).
- **Advanced RAG & Vector Search:** Retrieval-Augmented Generation (RAG), Semantic Chunking, Vector Embeddings, PostgreSQL pgvector, Qdrant, ChromaDB, Hybrid Keyword/Dense Search, Semantic Caching (Redis).
- **Agent Evaluation & Observability:** Eval-Driven Development, RAGAS Benchmarking, Regression Detection Test Suites, Langfuse Tracing, OpenTelemetry, Token Budgeting, Latency Profiling, Automated Hallucination Guardrails.
- **Backend Engineering & APIs:** Python (Advanced - AsyncIO, FastAPI, PyTest, Typing), TypeScript, Node.js, REST APIs, WebSockets, Docker, Linux, Microservices Architecture.
- **Security & Engineering Best Practices:** Defensive Software Architecture, Secure Credential Isolation, Code Reviews, ADRs, Git, CI/CD (GitHub Actions), High-Reliability Deployments.

Professional Experience

AI Engineer & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote Europe)

- Architected and shipped production Generative AI platforms, internal agent frameworks, and advanced RAG systems for European startups, taking solutions from conceptual architecture to live deployment.
- Engineered custom Model Context Protocol (MCP) servers and clients, allowing autonomous agents to query databases, run analysis tools, and interact with external APIs through standardized interfaces.
- Built resilient multi-step agent orchestration loops featuring persistent state management, dynamic context trimming, and human-in-the-loop validation gates.
- Developed automated evaluation pipelines using RAGAS, establishing systematic benchmarks for agent decision quality, tool-calling precision, and retrieval relevance across prompt iterations.
- Integrated real-time distributed tracing with Langfuse and OpenTelemetry, providing full observability into token economics, latency bottlenecks, and agent reasoning traces.
- Packaged AI services into containerized Docker microservices and established automated CI/CD regression test suites via GitHub Actions.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Developed high-performance asynchronous Python and TypeScript microservices for automated document ingestion, entity extraction, and structured classification.
- Built vector indexing pipelines and caching infrastructure using PostgreSQL and Redis, ensuring sub-second query retrieval across large unstructured corpora.
- Implemented comprehensive unit and integration testing harnesses using PyTest, maintaining high code reliability across deployment cycles.
- Collaborated directly with technical founders and platform teams to translate complex product goals into resilient, modular backend architectures.

Key Engineering Projects

- **Autonomous Tool-Using Agent Platform (Python + LangGraph + MCP + Postgres):** Multi-agent collaborative execution framework executing verified tool operations via Model Context Protocol against enterprise databases, featuring state preservation and Langfuse tracing.
- **Advanced Document Intelligence & RAG Engine (FastAPI + pgvector + RAGAS):** End-to-end knowledge engine combining dense semantic retrieval, hybrid search, semantic caching, and automated regression evaluation harnesses.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous foundation in formal logic, decision systems, computational theory, and complex systems analysis.
- **Languages:** English (C1 Professional, fluent written/spoken), Italian (Working proficiency).
- **Work Authorization & Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted legal right to live and work across the European Union without sponsorship; available immediately for remote engagement.